Sequence Alignment

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Introduction

- Sequence alignment is a way of arranging the sequences of DNA, RNA, or protein to identify regions of similarity that may be a consequence of functional, structural, or evolutionary relationships between the sequences.
- Aligned sequences of nucleotide or amino acid residues are typically represented as rows within a matrix.
- Gaps are inserted between the residues so that identical or similar characters are aligned in successive columns.
- It is the procedure for comparing two or more sequences by searching for a series of individual characters or character patterns that are in the same order in the sequences.

Introduction

- ❑ To find whether two (or more) genes or proteins are evolutionarily related to each other.
- ❑ To find structurally or functionally similar regions within proteins.
- ❑ <u>Sequence alignment is important for:</u>
  - ➢ prediction of function
  - ➢ database searching
  - ➢ gene finding
  - ➢ sequence divergence
  - ➢ sequence assembly

Terminologies

- **Sequence identity:** exactly the same Amino Acid or Nucleotide in the same position.
- **Sequence similarity:** Substitutions with similar chemical properties (purines [A, G], pyrimidines [C, T]).
- **Sequence homology:**

  General term that indicates evolutionary relatedness among sequences; we usually measure of percentage identity of sequence homology
- **Pairwise alignment:**

  Used to find the best-matching piecewise (local) or global alignments of two query sequences. Pairwise alignments can only be used between two sequences at a time.
- **Multiple sequence alignment:**

  Try to align all of the sequences in a given query set.

Terminologies

- **Global Alignment:**

- In global alignment, two sequences to be aligned are assumed to be generally similar over their entire length.
- Alignment is carried out from beginning to end of both sequences to find the best possible alignment across the entire length between the two sequences.
- This method is more applicable for aligning two closely related sequences of roughly the same length.
- For divergent sequences and sequences of variable lengths, this method may not be able to generate optimal results because it fails to recognize highly similar local regions between the two sequences.

- **Local Alignment:**

- Local alignment, on the other hand, does not assume that the two sequences in question have similarity over the entire length.
- It only finds local regions with the highest level of similarity between the two sequences and aligns these regions without regard for the alignment of the rest of the sequence regions.
- This approach can be used for aligning more divergent sequences with the goal of searching for conserved patterns in DNA or protein sequences.

Classification

I. <u>Pairwise:</u>

a. Global:

- *Algorithm:*
  - Needleman Wunsch algorithm is used in software used for alignment
- *Softwares/Tools:*
  - **Emboss Needle:** creates an optimal global alignment of two sequences
  - **Emboss Stretcher:** uses a modification of the Needleman-Wunsch algorithm that allows larger sequences to be globally aligned
  - **GGSEARCH2SEQ:** finds an optimal global alignment
  - **BioPython:** the BioPython library offers pairwise sequence alignment capabilities. SeqIO module can be used for sequence alignment by using the pairwise2 function for global alignments.

b. Local:

- *Algorithm:*
  - Smith-Waterman
- *Software/Tools:*
  - Emboss Water, Emboss Matcher, FASTA, BLAST (used in both pairwise and multiple sequence alignment to find out similarity)

Classification

II. Multiple:

- *Softwares:*

a. Global: Clustal Omega, MAFFT, Muscle, T-Coffee
b. Local: Kalign, M-View(colored to identify similarity, conserved regions, gap)

All MSA tools help in finding similarity but use different algorithms

- *Algorithm for phylogeny (Post MSA):*
  - UPGMA, Neighbor-Joining, Maximum Likelihood

Scoring Matrix (Used for both Pairwise and Multiple sequence alignment):
PAM, BLOSUM

Tools of MSA

Clustal OMEGA

- ✓ Clustal Omega is a multiple sequence alignment program that uses seeded guide trees and HMM profile-profile techniques to generate alignments between three or more sequences.
- ✓ It produces biologically meaningful multiple sequence alignments of divergent sequences. Evolutionary relationships can be seen via viewing Cladograms or Phylograms.

KALIGN

- ✓ A fast and accurate multiple sequence alignment algorithm for DNA and protein sequences.
- ✓ This tool uses the Wu-Manber string-matching algorithm, to improve both the accuracy and speed of multiple sequence alignment.
- ✓ It is a fast and robust alignment method, especially well-suited for the increasingly important task of aligning large numbers of sequences.

Tools of MSA

MAFFT

- ✓ MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program which implements the Fast Fourier Transform (FFT) to optimise protein alignments based on the physical properties of the amino acids.
- ✓ The program uses progressive alignment and iterative alignment. MAFFT is useful for hard-to-align sequences such as those containing large gaps (e.g., rRNA sequences containing variable loop regions).

MUSCLE

- ✓ MUSCLE stands for Multiple Sequence Comparison by Log-Expectation. MUSCLE is claimed to achieve both better average accuracy and better speed than ClustalW2 or T-Coffee, depending on the chosen options.
- ✓ MUSCLE enables high-throughput applications to achieve average accuracy comparable to the most accurate tools previously available, which is expected to be increasingly important in view of the continuing rapid growth in sequence data.
- ✓ Multiple alignments of protein sequences are important in many applications, including phylogenetic tree estimation, secondary structure prediction and critical residue identification.